

Financial Literacy and Food Insecurity Among Seniors in Clark County

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Executive Summary

This report looks at whether financial literacy helps explain food insecurity among senior households in Clark County. Using data from the 2019 Clark County Senior Food Insecurity Survey, the analysis looks at both food insecurity overall and very low food security, the most severe form of hardship.

The results show a clear difference between the two outcomes. Before accounting for other household characteristics, stronger financial literacy is linked to a lower likelihood of both broad food insecurity and very low food insecurity. Once factors such as income, education, health, age, household composition, gender, and housing tenure are taken into account, however, the relationship with broad food insecurity becomes much weaker and is no longer statistically significant. The relationship with very low food security also becomes smaller, but it remains negative and statistically significant.

The main takeaway is that financial literacy does not appear to broadly explain senior food insecurity. Instead, the relationship is more consistent at the most severe level of hardship. In the preferred model, a one-unit increase in the financial literacy index is associated with a 5.5 percentage point lower probability of very low food security. That general result also holds up across several alternative measures and models.

These findings should be interpreted carefully. The study shows associations, not cause and effect, and cannot determine whether financial literacy reduces severe hardship or whether severe hardship makes financial management more difficult.

For stakeholders, financial education should not be treated as a standalone solution. Income, health, housing, and access to support remain central. Still, the persistent relationship with very low food security may be worth testing through small-scale efforts that combine financial capability support with existing services.

Introduction and Background

Throughout the United States, many older adults face food insecurity, a serious nutrition-related public health issue. The USDA defines food-insecure households as those that, at times during the year, are uncertain of having or unable to acquire enough food because of insufficient money or other resources. This can include worrying about running out of food, reducing meal quality, cutting meal size, or skipping meals.

In 2024, 13.7 percent of U.S. households, or 18.3 million households, experienced food insecurity at some point during the year (USDA ERS, 2026). This prevalence shows that food insecurity remains a major national issue, especially for older adults who may face fixed incomes, medical costs, transportation barriers, and limited access to support services. For older adults, these challenges are closely tied to well-being because food insecurity can overlap with health needs, independence, financial stress, and the ability to maintain stable daily living conditions.

Early assessments often treated poverty as the primary determinant of food insecurity. However, more recent research has shown that many households above the poverty line still experience food insecurity, while many households below the poverty line do not. This gap has led researchers to consider other household-level factors, including financial literacy and financial management. Millimet et al. highlight this concern directly, writing that “inadequate financial skills and practices are a significant problem that spans all socioeconomic groups” (Millimet et al., 2015).

Research on older adults shows that food insecurity is tied to more than income alone. Seniors may face health problems, medical expenses, housing costs, limited mobility, and other competing demands that make it harder to consistently afford enough food. Recent work on older adult food insecurity emphasizes how these pressures can overlap, forcing some households to make difficult tradeoffs between food and other basic needs. This makes food insecurity among seniors a broader household-

resource problem rather than simply a reflection of whether income falls above or below a particular threshold (Heflin and Harrington Meyer, 2025).

Research on financial literacy points to another part of that picture. Millimet et al. (2015) find that financial skills and practices may help explain differences in food security even among the households facing serious economic vulnerability. This suggests that the way households manage limited resources could matter alongside the amount of resources they have. However, less is known about whether that relationship looks the same across different levels of food hardship, particularly among seniors. The current study builds on that question by examining whether financial literacy is related to food insecurity overall, the most severe form of food hardship, or both.

Financial literacy is relevant to this issue because food insecurity is not determined by income alone. While limited resources do have an impact, households also face decisions about budgeting, debt, savings, benefit use, and tradeoffs between food, housing, medical care, and transportation. For seniors, the difficulty of these decisions may be exacerbated due to health-related issues and living on fixed incomes. As a result, financial literacy may help explain why some households with similar economic constraints experience different levels of food hardship.

This report examines whether financial literacy helps explain food insecurity among senior households in Clark County. More specifically, the analysis examines the role that financial literacy plays in explaining the probability that a senior household could be classified as food insecure. Because the severity of food insecurity can vary, the report considers both broad food insecurity and very low food security. The key question is whether financial literacy helps explain general food hardship, the most severe form of food hardship, or both.

Empirics

Data and Sample

The analysis uses survey data from the Clark County Senior Food Insecurity Survey, which was designed to better understand senior hunger and access to food in Southern Nevada. The survey targeted households with at least one individual age 60 or older living in Southern Nevada and was sponsored by Three Square Food Bank. Respondents were asked about household food access, demographics, income, health, housing, social support, SNAP participation, spending tradeoffs, neighborhood conditions, and financial literacy (UNLV School of Public Health, 2019).

The original dataset contains 800 observations and 112 variables. During the data audit, 245 exact duplicate rows were identified, leaving 555 unique observations after deduplication. The repeated rows were identical across all 112 variables, including the household identifiers, and therefore do not provide independent information. The deduplicated sample is used as the preferred analytic sample, while the full sample is retained as a robustness check to verify that the results are not sensitive to the duplication-removal process.

Because the study focuses on senior food insecurity, the sample should not be interpreted as representative of all Clark County seniors. In addition to screening for eligible senior households, the survey included higher-risk oversampling based on household income. Among respondents who reported household income, approximately 31 percent were classified as broadly food insecure, indicating that the sample includes a substantial share of households experiencing food hardship.

The survey focused on senior households. Screening questions verified whether an individual age 60 or older lived in the household, whether the respondent was age 60 or older or could answer for the senior household member, and whether the household lived in Clark County, Nevada. The survey also included

an additional screening question for higher-risk oversampling based on whether annual household income exceeded \$60,000 (Clark County Senior FIS Datamap, 2019).

The survey collected information across several areas related to senior well-being, including demographics, income, household composition, food spending, food security, health, housing, social support, SNAP participation, spending tradeoffs, neighborhood access, and financial literacy. For this analysis, the most important components are the food security questions, financial literacy questions, and the household characteristics used as controls.

The food security questions are used to construct the outcome variables, while the financial literacy questions are used to construct the primary explanatory variable. Additional household characteristics are later incorporated as controls to account for observed factors that may be related to both financial literacy and food insecurity.

Outcome Variables

The analysis uses two binary outcome variables based on the USDA six-item food security module included in the survey (questions 17-22). Each outcome is coded as 1 if the household meets the relevant food insecurity classification and 0 otherwise. The first outcome is broad food insecurity, which identifies households classified as either low food security or very low food security. The second outcome is very low food security, which captures the most severe level of food hardship. This distinction is important because very low food security is nested within broad food insecurity. In other words, all households classified as very low food security are also food insecure, but not all food-insecure households experience very low food security. Using both outcomes allows the analysis to test whether financial literacy is associated with general food hardship, severe food hardship, or both.

Following USDA guidance for the six-item module, affirmative responses are summed to create a raw food security score. Households with raw scores of 2 through 4 are classified as having low food security, while households with raw scores of 5 or 6 are classified as having very low food security. For the broad food insecurity outcome, households with either low or very low food security are coded as food insecure. For consistency with the analysis, the broad food insecurity outcome is labeled `food\_insecure`, while the severe food hardship outcome is labeled `very\_low\_security` (USDA ERS, 2024).

Financial Literacy Measure

The key explanatory variable in the analysis is financial literacy. Because financial literacy is not measured by a single survey question, the analysis uses several items related to current financial confidence and day-to-day financial behavior. These items capture related dimensions of financial capability, including confidence managing financial matters, budgeting, tracking expenses, reviewing bills, paying bills on time, finding ways to reduce expenses, and reviewing income and expenses before large purchases.

The main financial literacy measure is constructed from the respondent's current financial confidence question, q44, and eight financial behavior questions, q45a through q45h. The q44 item asks respondents how confident they are in dealing with day-to-day financial matters, such as checking accounts, credit and debit cards, and tracking expenses. The q45 items ask about financial practices such as setting financial goals, using a spending plan or budget, keeping track of spending, reviewing bills for accuracy, paying bills on time, paying bills using a checking account, finding ways to decrease expenses, and reviewing income and expenses before making large purchases (Clark County Senior FIS Datamap, 2019).

The primary financial literacy measure is a standardized z-score index. Each component was standardized by subtracting its sample mean and dividing by its sample standard deviation. This places the items on a common scale, so that variables measured on different response scales can be combined. The standardized items were then averaged across the included financial literacy components to create `finlit\_z\_index`, with higher values representing stronger measured financial literacy or financial management behavior. This index is preferred because it is transparent, easy to interpret, and does not rely on an arbitrary cutoff or any single survey question.

As a robustness check, the analysis also uses an additional indexing technique called principal component analysis, or PCA. PCA creates an alternative summary measure by identifying the common variation across the financial literacy items. The first principal component was used because it captures the largest amount of shared variation across the included items. In this analysis, the first component explained approximately 43% of the standardized variation across the nine items and had an eigenvalue of approximately 3.9, indicating that it captured substantially more variation than any single standardized item. The resulting PCA-based measure is labeled `finlit\_pca\_index`. Using both `finlit\_z\_index` and `finlit\_pca\_index` allows the analysis to test whether the results are sensitive to the way financial literacy is measured. The nine items included in both financial literacy indexes contained no missing observations in either analysis sample, so index construction did not reduce available sample size.

One related survey item, q43, asks respondents to recall their parent or guardian's confidence with day-to-day financial matters during childhood. This item was not included in the preferred financial literacy index because it reflects childhood family background rather than the respondent's current financial literacy or current financial management behavior. Excluding q43 helps keep the index focused on the respondent's own present-day financial capability.

Control Variables and Data Preparation

The relationship between financial literacy and food insecurity may be confounded by other household characteristics that are associated with both. Income is a clear example. Households with greater financial resources may also report stronger financial literacy and, independently, face a lower risk of food insecurity. Similar concerns apply to education, health, household composition, and other demographic factors. To help account for observed confounding, the controlled specifications include seven covariates: age, education, household income, number of other household members, self-rated health, gender, and housing tenure.

Before estimation, each control variable was reviewed for coding consistency and missing or non-substantive responses. This process revealed several issues that required correction before the variables could be incorporated into the models. The most consequential issue involved the number of other household members. Respondents living alone were not asked the survey follow-up question on the number of other household members, leaving their responses blank. These blanks therefore represented a skip pattern rather than true missing data. This interpretation was verified against a separate indicator identifying respondents who lived alone, and the affected observations were recoded as zero additional household members rather than excluded from the estimation sample.

The remaining controls were cleaned using the same general principle: response categories such as ‘don’t know’ and ‘refused’ were treated as missing rather than as substantive numeric values. Invalid or non-substantive response codes were similarly excluded from the ordered education, income, and self-rated health measures, while gender was coded as a binary indicator for use in the regression models.

Supplemental Nutrition Assistance Program (SNAP) participation was not included in the primary control set because participation may be endogenous to food hardship. Households may enroll in SNAP in response to financial strain or food insecurity, while SNAP benefits may also directly affect food access.

Treating participation as a standard control could therefore condition on a variable that lies partly within the process generating the outcome. For this reason, SNAP participation was excluded from the preferred specification.

Several controls, including household income, education, and self-rated health, are measured using ordered response categories. An initial specification entered these variables as full sets of category indicators. However, some categories contained relatively few observations, producing sparse cells and unstable logistic regression estimates. The preferred specification therefore enters these variables as ordered covariates rather than as separate indicators for each category. Housing tenure, which has no natural ordering, remains categorical.

This specification improves model stability and preserves a consistent control structure across the linear probability and logistic models. The tradeoff is that it imposes a stronger functional-form assumption: moving from one category to the next is assumed to have a common incremental relationship with the outcome.

Econometric Approach

The relationship between financial literacy and food insecurity is estimated using both a linear probability model and logistic regression. Each model is estimated separately for the broad food insecurity outcome and the very low food security outcome. The preferred specification uses the deduplicated sample and the standardized z-score financial literacy index, while the full sample and PCA-based index are used as robustness checks.

The general specification can be written as:

where y_{it} represents either broad food insecurity or very low food security for household i , x_{it} is the household's financial literacy index, and z_{it} is the vector of control variables described in the previous subsection. The constant term is represented by α , while β represents the vector of coefficients associated with the control variables. The coefficient of primary interest is γ , which captures the conditional association between financial literacy and the probability of food insecurity after accounting for the included household characteristics. The error term, ϵ_{it} , captures other unobserved factors affecting food insecurity that are not included in the model.

The linear probability model provides a direct estimate of the change in the probability of the outcome associated with a one-unit increase in the financial literacy index. Because both outcomes are binary, logistic regression is also estimated to account for the nonlinear relationship between the covariates and predicted probabilities. To make the logistic estimates more directly comparable with the linear probability model, the results are reported as average marginal effects rather than log-odds coefficients. For the linear probability models, HC1 heteroskedasticity-robust standard errors are used for statistical inference.

Both baseline and controlled specifications are estimated. The baseline models include financial literacy as the sole explanatory variable and provide an initial measure of the unadjusted association. The controlled models then add age, education, household income, number of other household members, self-rated health, gender, and housing tenure. Comparing the two specifications shows how the estimated relationship changes after accounting for observed household characteristics that may be related to both financial literacy and food insecurity.

Model Assumptions and Diagnostic Checks

The credibility of the estimated relationships depends on several assumptions about the specification and the data. Most importantly, the models assume that the included covariates and chosen functional

forms provide an adequate specification of the conditional relationship between financial literacy and food insecurity. The linear probability model also assumes a linear relationship between the included covariates and the probability of the outcome, while the logistic model imposes a logistic functional form. As discussed above, the treatment of household income, education, and self-rated health as ordered covariates further assumes that movement between adjacent response categories has a common incremental relationship with the outcome.

These assumptions support interpretation of the estimated associations, but they do not establish causality. A causal interpretation would require financial literacy to be uncorrelated with unobserved determinants of food insecurity contained in the error term. Because the data are observational and cross-sectional, unmeasured characteristics may affect both financial literacy and food insecurity, and the direction of influence may also run partly from food hardship to financial behavior. These limitations are addressed more fully in the Limitations and Next Steps section.

The models were also evaluated for multicollinearity among the included control variables. Variance inflation factors were examined to determine whether strong linear relationships among the covariates were inflating coefficient uncertainty or making the estimates unstable. The diagnostic results did not indicate substantial multicollinearity among the included controls.

The controlled models are estimated using complete cases for the included covariates, which means the observations with missing values on any required variable are excluded from estimation. Income nonresponse accounts for an important share of these exclusions. A comparison of respondents who reported household income with those who did not showed that the two groups differed meaningfully in food insecurity prevalence. Approximately 13 percent of income nonrespondents were food insecure, compared with about 31 percent of respondents who reported income. The resulting complete-case sample is therefore not a random subset of the original survey and is somewhat more concentrated

among food-insecure households. This pattern does not invalidate the estimates, but it should be considered when interpreting their representativeness.

Results

Main Regression Results

The main results are presented for the preferred deduplicated sample using the standardized z-score financial literacy index. Baseline models first estimate the unadjusted relationship between financial literacy and each food insecurity outcome. Controlled models then account for age, education, household income, number of other household members, self-rated health, gender, and housing tenure. Results are reported using both the linear probability model and logit average marginal effects.

Across both outcomes, the baseline estimates indicate a negative association between financial literacy and food insecurity. The relationship changes after the control variables are introduced, however, and the pattern differs substantially by outcome. For broad food insecurity, the estimated association becomes smaller and is not statistically significant in the controlled models. For very low food security, the association also attenuates after adjustment but remains negative and statistically significant under both estimators.

Table 1. Main Regression Results, Deduplicated Sample and Z-Score Financial Literacy Index

Outcome	Model	Estimate / AME	p-value	N
Broad food insecurity	Baseline LPM	-0.158	<0.001	555
Broad food insecurity	Controlled LPM	-0.052	0.108	490
Broad food insecurity	Baseline Logit AME	-0.143	<0.001	555
Broad food insecurity	Controlled Logit AME	-0.048	0.084	490
Very low food security	Baseline LPM	-0.101	<0.001	555
Very low food security	Controlled LPM	-0.061	0.017	490
Very low food security	Baseline Logit AME	-0.079	<0.001	555
Very low food security	Controlled Logit AME	-0.055	0.003	490

Note: LPM = Linear Probability Model; AME = Average Marginal Effect. Controlled models include age, education, household income, number of other household members, self-rated health, gender, and housing tenure. HC1 heteroskedasticity-robust standard errors are used for the LPMs.

Broad Food Insecurity

Financial literacy is strongly and negatively associated with broad food insecurity in the baseline models.

In the linear probability model, a one-unit increase in the z-score financial literacy index is associated with a 15.8 percentage point lower probability of broad food insecurity ($p < 0.001$). The baseline logit model produces a similar average marginal effect, corresponding to a 14.3 percentage point lower probability of broad food insecurity ($p < 0.001$).

Once the full set of controls is added, however, the estimated relationship becomes substantially smaller. The controlled linear probability model produces an estimate of -0.052 ($p = 0.108$), while the controlled logit average marginal effect is -0.048 ($p = 0.084$). Although both estimates remain negative, neither is statistically significant at the 5% level.

This attenuation suggests that part of the strong baseline relationship reflects observed household characteristics that are associated with both financial literacy and food insecurity. After accounting for age, education, household income, number of other household members, self-rated health, gender, and housing tenure, the remaining association between financial literacy and broad food insecurity is not estimated precisely enough to distinguish it from zero at conventional significance levels.

Very Low Food Security

Financial literacy is also strongly and negatively associated with very low food security in the baseline models. In the linear probability model, a one-unit increase in the z-score financial literacy index is associated with a 10.1 percentage point lower probability of very low food security ($p < 0.001$). The baseline logit model produces an average marginal effect corresponding to a 7.9 percentage point lower probability of very low food security ($p < 0.001$).

Unlike the results for broad food insecurity, the association remains statistically significant after the control variables are added. The controlled linear probability model produces an estimate of -0.061 ($p = 0.017$), while the controlled logit average marginal effect is -0.055 ($p = 0.003$). Under the preferred logit specification, a one-unit increase in the financial literacy index is therefore associated with a 5.5 percentage point lower probability of experiencing very low food security, conditional on the included household characteristics.

The similarity between the controlled linear probability and logit estimates strengthens the result.

Although the magnitude of the association is smaller than in the baseline models, it remains negative, statistically significant, and broadly consistent across both estimators. This contrasts with the broad food insecurity results, for which the controlled estimates are not statistically significant.

Robustness Checks

The main pattern is not limited to a single estimator, financial literacy measure, or analysis sample. To assess the sensitivity of the results, the controlled models were re-estimated using both the z-score and PCA-based financial literacy indexes, both the deduplicated and full samples, and both the linear probability and logit specifications. Because the z-score and PCA indexes are measured on different scales, their coefficient magnitudes are not directly comparable. The robustness assessment therefore focuses primarily on the direction of the estimates, statistical significance, and consistency of the overall pattern.

For broad food insecurity, the controlled estimates remain generally negative but are not statistically significant across most specifications. In the deduplicated sample, neither the z-score nor PCA-based estimate is statistically significant under either estimator. The same general pattern holds in the full sample, where the estimates are close to zero and statistically insignificant. Taken together, these results reinforce the conclusion that the negative baseline association between financial literacy and broad food insecurity does not remain statistically robust after the inclusion of household controls.

The results for very low food security are more consistent. In the deduplicated sample, the association is negative and statistically significant under both financial literacy measures and both estimators. Using the z-score index, the controlled estimate is -0.061 in the linear probability model ($p = 0.017$) and the logit average marginal effect is -0.055 ($p = 0.003$). Using the PCA-based index, the corresponding estimates are -0.019 ($p = 0.025$) and -0.017 ($p = 0.006$). The smaller PCA coefficients reflect the different scale of that index rather than evidence of a weaker underlying association.

The full sample results are somewhat less uniform but continue to support the same broad pattern. The estimated relationship with very low food security remains negative across all specifications. Under the logit model, both financial literacy measures produce statistically significant average marginal effects,

while the linear probability estimates are weaker. Overall, the severe-outcome result is substantially more persistent across reasonable analytical choices than the result for broad food insecurity.

Table 2. Robustness Checks, Controlled Logit Average Marginal Effects

Outcome	Financial Literacy Measure	Sample	AME	p-value	N
Broad Food Insecurity	z-score index	Deduplicated	-0.048	0.084	490
Broad Food Insecurity	z-score index	Full	0.001	0.981	712
Broad Food Insecurity	PCA index	Deduplicated	-0.014	0.132	490
Broad Food Insecurity	PCA index	Full	0.003	0.749	712
Very Low Food Security	z-score index	Deduplicated	-0.055	0.003	490
Very Low Food Security	z-score index	Full	-0.038	0.017	712
Very Low Food Security	PCA index	Deduplicated	-0.017	0.006	490
Very Low Food Security	PCA index	Full	-0.011	0.034	712

Note: AME = Average Marginal Effect. All models include age, education, household income, number of other household members, self-rated health, gender, and housing tenure. The z-score index is the primary financial literacy measure, while the PCA index is used as a robustness check.

Subgroup Vulnerability

The models account for several demographic and socioeconomic characteristics that may be related to food insecurity, including age, education, household income, number of other household members, self-rated health, gender, and housing tenure. However, the analysis does not estimate the separate financial literacy effects across these groups or test interaction effects between financial literacy and individual household characteristics. The results therefore do not identify whether the relationship between financial literacy and food insecurity is systematically stronger for particular subgroups of seniors.

What the analysis does show is that the relationship differs by the severity of food hardship. Financial literacy is not robustly associated with broad food insecurity after controls are added, but it remains

more consistently associated with very low food security. This distinction concerns the severity of the outcome rather than differential effects across demographic or socioeconomic subgroups.

Taken together, the results provide a clear answer to the primary research question. Financial literacy is strongly associated with both broad and severe food insecurity in the baseline models, but the relationship changes once observed household characteristics are considered. For broad food insecurity, the adjusted estimates are smaller and not statistically significant. For very low food security, the association remains negative, statistically significant, and comparatively robust across estimators, financial literacy measures, and analysis samples. The evidence therefore points to a more persistent relationship between financial literacy and the most severe form of food hardship than with food insecurity generally.

Discussion

The central finding of this study is not simply that financial literacy is related to food insecurity. Rather, the relationship changes depending on the severity of the hardship being examined. Financial literacy is strongly associated with both food insecurity outcomes before household characteristics are taken into account. After adjusting for income, education, health, and other observed factors, however, the relationship with broad food insecurity becomes smaller and is no longer statistically significant. The relationship with very low food security is more persistent, remaining negative and statistically significant in the preferred models and across several alternative specifications.

One of the more surprising results was how differently the two outcomes behaved after controls were added. At the outset, it would have been reasonable to expect financial literacy to relate to food insecurity more generally. Instead, the findings suggest that financial literacy may be less important in explaining whether a household crosses the broad food insecurity threshold than in understanding the most severe level of food hardship. General food insecurity may be driven more heavily by structural and

resource constraints, such as insufficient income, poor health, and housing costs, that financial skills alone cannot easily offset. For very low food security, however, financial confidence and day-to-day financial behavior may matter more because households are operating under much tighter constraints.

This interpretation should remain cautious. The results do not show that stronger financial literacy prevents seniors from experiencing very low food security. It is also possible that severe food hardship makes financial planning and management more difficult, or that other unmeasured characteristics influence both financial literacy and food security. The findings are therefore best understood as evidence of a more persistent association between financial literacy and severe food hardship, not proof of a causal effect.

The results partly line up with earlier research on financial literacy and food insecurity. Millimet et al. (2015) argue that financial skills and practices may help explain food security outcomes beyond income alone. Our findings point in the same general direction, but the relationship is more limited than we initially expected. Once household characteristics are taken into account, financial literacy is no longer significantly associated with broad food insecurity. The relationship is more persistent for very low food security, suggesting that financial literacy may matter more for the severity of hardship than for whether a household becomes food insecure in the first place.

That difference adds an important nuance to the existing research. Rather than finding the same relationship across all levels of food insecurity, our results suggest that the association may be concentrated at the most severe end of the spectrum. At the same time, the study cannot tell us whether financial literacy actually prevents households from moving into more severe hardship, so this pattern should still be interpreted cautiously.

For community organizations and other stakeholders, the main implication is that financial literacy should not be viewed as a simple explanation for senior food insecurity. The broad food insecurity result

weakens considerably once income, education, health, and other household characteristics are taken into account. That suggests the larger problem is closely tied to the economic and structural conditions seniors face, not just to how well they manage money. Financial skills may matter, but they are unlikely to substitute for adequate income, stable housing, access to food assistance, or healthcare.

At the same time, the more persistent relationship with very low food security is still important. The result suggests that financial confidence and day-to-day financial behavior may be especially relevant when households are under the greatest strain. For organizations working directly with seniors, that makes severe food hardship an important area for closer attention. The finding does not tell us that financial education will reduce hunger, but it does suggest that the connection between financial capability and the most extreme food hardship deserves further investigation.

Taken together, the study adds a more nuanced view of the relationship between financial literacy and food insecurity among seniors. The findings do not support the idea that financial literacy is a broad explanation for food insecurity, but they do show a more persistent relationship with the most severe form of hardship. That distinction matters because it suggests that the role of financial capability may depend on where a household falls along the food security spectrum. The study therefore contributes less by identifying a simple solution and more by narrowing the question: financial literacy may be most relevant when hardship is already severe, and that is where future research should look more closely.

Recommendations

The findings do not support broad claims that financial literacy programs will reduce food insecurity among seniors. The study identifies associations rather than causal effects, and it does not evaluate any specific intervention. The recommendations that follow are therefore intended to identify where stakeholders may want to focus attention, further investigation, and careful evaluation rather than to prescribe a proven solution.

Do Not Treat Financial Literacy as a Standalone Solution

The results do not support financial literacy as a broad explanation for senior food insecurity. Once income, education, health, and other household characteristics are considered, the relationship with broad food insecurity becomes smaller and is no longer statistically significant. For stakeholders, that means financial capability should not be treated as a substitute for more fundamental forms of support related to income, food access, healthcare, and housing stability.

Financial skills may still be useful, but the findings suggest they are only one part of a much larger picture. Programs or policies that focus narrowly on budgeting or financial education risk placing too much emphasis on individual behavior while overlooking the structural conditions that contribute to food hardship. Any financial capability effort should therefore be considered alongside, rather than in place of broader supports for seniors facing economic insecurity.

Use the Severe-Food-Hardship Finding to Guide Further Testing

The more persistent relationship with very low food security deserves attention, but it should not be treated as proof that a financial capability program would reduce severe food hardship. A better next step would be to use this finding to guide targeted follow-up. Community organizations working with seniors experiencing the most severe forms of food insecurity could consider small-scale pilot efforts that combine financial capability support with existing food, income, or benefit services. Those efforts should be evaluated from the outset to determine whether food security outcomes actually improve over time.

Invest in Stronger Research Before Scaling Programs

The current study narrows the question, but it cannot answer the causal one. Future research should focus on whether changes in financial literacy or financial behavior actually lead to changes in food

security. Longitudinal data that follow the same households over time would help establish whether changes in financial capability tend to occur before changes in food hardship. Evaluations of actual programs would provide even stronger evidence.

Until that evidence exists, stakeholders should be cautious about scaling financial literacy initiatives on the assumption that they will reduce food insecurity. A more defensible approach is to test these ideas on a small scale, measure the outcomes, and refine programs before expanding them. If future evaluations show that financial capability support improves food security outcomes, especially among seniors experiencing severe hardship, those results would provide a much stronger basis for larger investments and broader implementation.

Limitations and Next Steps

The most important limitation of this study is that the data are observational and cross-sectional. Financial literacy and food security were measured at the same point in time, so the analysis can identify associations but cannot determine the direction of causality. Unmeasured characteristics may influence both financial literacy and food insecurity, and severe food hardship may itself make financial planning and management more difficult. Adding controls helps account for observed differences between households, but it does not eliminate these identification concerns.

The financial literacy measure also has limitations. It is constructed from self-reported confidence and financial behaviors rather than from a direct objective test of financial knowledge. Responses may therefore contain measurement error or reflect how respondents perceive and report their own behavior. Although the z-score and PCA-based indexes produce highly consistent results, both are built from the same underlying survey items and therefore share these measurement constraints.

A third limitation involves the estimation sample. The controlled models use complete cases, and income nonresponse accounts for an important share of the excluded observations. Respondents who did not report income had substantially lower food insecurity rates than those who did, meaning the final estimation sample is not a random subset of the original survey. As discussed in the diagnostic section, this pattern may limit the representativeness of the controlled estimates and should be kept in mind when interpreting results.

The age and geographic scope of the survey also limit how broadly the findings should be applied. The data were collected in 2019, before the economic and public health disruptions of the following years, and the relationship between financial literacy and food insecurity may look different under current conditions. In addition, the survey focuses on senior households in Clark County and was not designed to represent older adults everywhere. The findings are therefore most directly relevant to the population studied and should be generalized to other communities with caution.

These limitations point to several clear next steps. The most useful improvement would be data that follow the same households over time. Longitudinal data would make it possible to observe whether changes in financial literacy tend to occur before changes in food security and would help separate the effects of stable personal characteristics that cannot be measured in the current cross-sectional design.

A second priority is the evaluation of actual financial capability programs. Rather than assuming that the observed association will translate into an intervention effect, future studies could compare food security outcomes before and after participation and, where possible, against a similar comparison group. Randomized or quasi-experimental designs would provide stronger evidence about whether improvements in financial capability lead to changes in severe hardship.

Future work should also collect more recent and representative local data and examine whether the relationship differs across groups of seniors. The current analysis controls for age, income, health,

gender, housing tenure, and other characteristics, but it does not estimate whether the association between financial literacy and food insecurity is stronger for some groups than others. Larger samples would make those subgroup comparisons more feasible and could help identify where further attention is most warranted

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